Abdullah Al Zahid

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SUMMARY

Transportation engineer with a master's degree and part way through a computer science degree, currently an Assistant Planner with the City of Toronto. Builds and maintains spatial datasets and the tooling around them using GIS and Python, brings several sources into one validated dataset, and documents the result for the analysts who rely on it.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Build and maintain the road and right-of-way datasets the division depends on, using GIS and Python, including a ROW-width dataset covering all city roads, built with QGIS, MapBox and JavaScript.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.
- Build and maintain the datasets and tooling Transportation Planning depends on, using Python and GIS.
- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Write the briefing notes, memos and presentations that carry issues, options and recommendations to senior staff across divisions.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.
- Ran OpenTrack simulations of GO rail services to establish what proposed operating and infrastructure changes would do to run times, reliability and recovery before they were taken forward.
- Worked with schedulers, operations staff and engineers to settle the variables a service plan had to be built on.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Worked with large travel datasets and wrote up results for readers without a modelling background.

SKILLS

Spatial data and GIS: ArcGIS Pro, QGIS, MapBox. Python (Pandas, NumPy), SQL and JavaScript. Dataset validation and documentation. Dashboards and data visualization. Advanced Excel. Reports, briefing notes and presentations. MS Office.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018

BSc, Computer Science — University of British Columbia, in progress (2025–2027)